

noise or interference to another device or electronic system is known as jamming. This condition is created through intentional emission or reflection of electromagnetic energy or signals to distract the electromagnetic signals of another system.

The ability to effectively jam signal or group of signals typically requires the use of a high-power transmitter, which produces a signal that is stronger than the target signal. The susceptibility of the receiver is also an important factor in the amount of transmission power necessary to effectively jam or disrupt a communication link or system. The electromagnetic interference has the capability of disrupting radio communications, radar, and radio navigation systems.

Effects of EMI

EMI may affect the devices, causing malfunction or failure, which might have very serious consequences. Under certain circumstances, EMI also causes health hazards such as insomnia, nervousness, headache, leukemia, and miscarriages due to continuous exposure to EM fields and pulses.

There are numerous factors that affect radio wave propagation, which include transmission power, antenna gain, attenuation factors, and noise. In addition, there are ionospheric and tropospheric conditions that also have a dramatic effect on the propagation of radio transmission through free-space.

EMC Laboratories in India

GOVERNMENT LABS

1. Aeronautical Development Establishment (ADE), Bangalore for microwave measurement (DRDO)
2. Centre for Airborne Systems (CABS), Bangalore for electrical and electronics testing (DRDO)
3. Combat Vehicle Research & Development Establishment (CVRDE), Chennai for combat vehicle and battle tanks (DRDO)
4. Electrical Research and Development Association (ERDA), Gujarat
5. Electronics and Quality Development Centre, Gujarat
6. Electronics and Radar Development Establishment (LRDE), Bangalore (DRDO)
7. EMC design consultancy group, SAMEER, Kolkata (MeitY)
8. EMC Division, SAMEER - Centre for electromagnetics, Chennai (MeitY)
9. EMC Division, SAMEER - Centre for microwave research, Mumbai (MeitY)
10. EMC test lab and centre, quality assurance division, Bharat Electronics Ltd (Defence PSU)
11. EMI/EMC test facility, CPRI, Bangalore
12. EMI/EMC test facility, Electronics Corporation of India Limited (A DAE enterprise)
13. Global Automotive Research Centre (GARC) - NATRIP, Chennai for automobile sectors
14. National Aerospace Laboratories (NAL), Bangalore for adaptive antennas, FSS structures and materials (CSIR)
15. Research Centre Imarat (RCI), Hyderabad for electrical and electromagnetic testing of missile components (DRDO)
16. STQC-Electronics Test & Development Centre (MeitY)

PRIVATE LABS (RECOGNISED)

17. Associated Electronics Research Foundation
18. Astute Labs Pvt Ltd
19. Electromagnetic Compatibility Consulting Science Academy (ECCSA), Chennai
20. EMC and Durability Lab, HCL Technologies Ltd
21. EMC Laboratory, Robert Bosch Engineering and Business Solutions Pvt Ltd
22. EMC Test Services - Astra Microwave Products Ltd
23. EMI/EMC Test Facility, Linkwell Telesystems Pvt Ltd
24. EMI/EMC Testing Laboratory, Emi Solutions Pvt Ltd
25. Genus Meter Testing Laboratory
26. Hi Physix Laboratory
27. Honeywell Technology Solutions, Bangalore - EMC, Env/Safety Lab
28. Labgo, Bangalore
29. Philips Lighting Laboratory, Philips Lighting India Ltd
30. Power Transformer Testing Laboratories, Transformers & Rectifiers (India) Ltd
31. Robert Bosch Engineering & Business Solutions India Ltd
32. SGS India Pvt Ltd, Electrical & Electronics Laboratory Consumer and Retail
33. Sigma Test and Research Centre, New Delhi
34. Tata Power Strategic Engineering Division, a division of the Tata Power Company Ltd
35. TUV SUD South Asia Pvt Ltd
36. UL India Lab, UL India Pvt Ltd
37. Yadav Measurements Pvt Ltd

PROFESSIONAL SOCIETY FOR EMC

38. Society of EMC Engineers (India)- www.semceindia.org
39. IEEE Electromagnetic Compatibility Society-<http://www.emcs.org>

TRAINING CENTRES FOR EMI/EMC

40. www.tonex.com
41. www.raceel.com/trainings/emi-emc-analysis
42. www.mindshare.com
43. <https://hollandshielding.com>
44. www.tuv.com
45. www.araiindia.com
46. www.lpci.in
47. www.scemcd.gov.in/train.php
48. www.cpri.in
49. www.tnqs.in
50. www.siepelindia.com
51. www.complus.in
52. www.emisindia.com